

## 46-929, Fall 2019: Homework #1

Due 5:30 PM, Wednesday, October 30.

Homework is submitted as a .pdf file on Canvas. Hand-written assignments are perfectly acceptable; of course, they should be clear.

Remember that you are to work **individually** on your homework assignments.

1. Let  $Z_t$ ,  $t = 0, \pm 1, \pm 2, \dots$  be independent normal random variables each with mean zero and variance  $\sigma^2$ , and let  $a$ ,  $b$ , and  $c$  be constants. Which, if any, of the following processes are stationary? Show all of your work. For each stationary process specify the mean and covariance function (i.e., specify  $E(X_t)$  and  $\gamma_X(h)$ ).

(a)  $X_t = a + bZ_t + cZ_{t-1}$

(b)  $X_t = a + Z_t^2$

(c)  $X_t = Z_t Z_{t-1}$

2. Let  $\{\epsilon_t\}$  be IID, mean zero white noise. Define for all  $t$

$$X_t = \epsilon_t + \epsilon_{t-1}\epsilon_{t-2}.$$

Show that  $\{X_t\}$  is (weak) white noise, but not an MDS.

3. Let  $\{Y_t\}$  be a stationary process with mean zero and autocovariance function  $\gamma_Y(h)$ . Let  $X_t = Y_t - Y_{t-1}$ . Show that  $\{X_t\}$  is stationary.
4. Choose any stock for which you can obtain data for the entirety of 2017. Construct the time series consisting of the **log returns** for this stock.
  - (a) Plot the time series. Is there clear evidence of non-stationarity?
  - (b) Construct the ACF of this series.
  - (c) Is there evidence that there is serial correlation in the series at any lag up to 20? Use the Ljung-Box test.

Include any plots and R code in your homework.